Experiment 7 - The Hall Effect

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October 21, 2026

1 Background

1.1 Discovery

In 1879, based on work by André-Marie Ampére and James Maxwell, Edwin Hall investigated how magnetic fields interacted with an electrical current in a conductor. He proposed that

$$E' \propto Mv$$

where $E' \equiv V$, $M \equiv |\vec{B}|$ and "v is the velocity of the electricity in the [object]" (E. H. Hall, 1879)[1] (At the time of Hall's paper the electron had only been theorized and wasn't discovered until 1897). Later, the equation was found to be

$$V_H = \frac{I_x B_z}{n_e C_e t}$$

where I_x is the current through the conductor, B_z is the magnetic field, n_e is the number of electrons, C_e is the charge of the electron and t is the thickness of the material. (Manny Soltero, 2023)[2]. The term $n_e C_e t$ represents the total charge density of the object.

The Hall effect describes how electrons accumulate on one side the material in the presence of a magnetic field.

1.2 Hall Effect Sensors

Hall effect sensors typically combine a Hall element and an op-amp. The Hall element has a similar structure to a Wheatstone bridge (R. Wei et al, 2017)[3]. The Hall voltage can be measured accross pins 1 and 3 when a bias voltage is applied between pins 2 and 4(Fig 1). If all of the resistors are not properly balanced the output voltage will be offset by some amount. In a larger IC the op-amp is used to amplify the voltage to levels that can be read by a computer.

Hall effect sensors are often used to measure the position or angle of something, for example a stick in a game controller, the rotation speed/postion of a motor's shaft or measuring current. Typically, magnets are mounted to one of the moving components and the sensor on the other side. If multiple magnets are used then the velocity of the object can be measured by maasuring the "bumpy" magnetic field. If the device itself can be calibrated or started from some known position then Hall effect sensors can be used to determine the position of the device to a comparitively high accuracy.

As Hall effect sensors are solid state components they can often be used in areas were mechanical wear of a more traditional component may be a problem, or where there might be a physical disconnect from the sensor and what it's trying to measure.

2 Method

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1. The apparatus was wired together. The voltmeter was chained in series with the top plug of the Hall effect waver, with the bottom of the wafer going back into the negative end of the power supply. The electromagnet was connected to the second power supply.

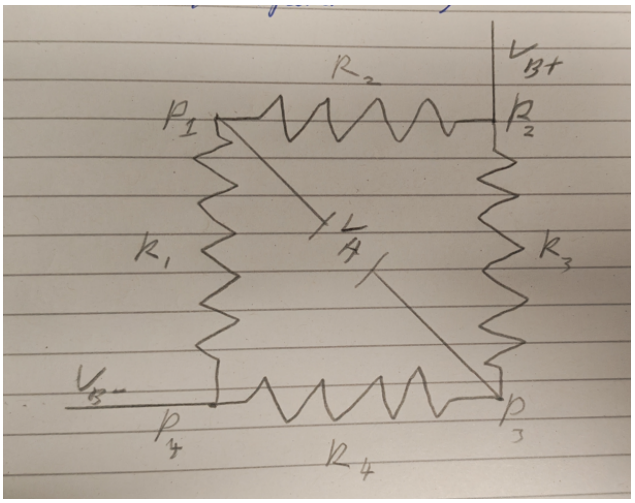


Figure 1: Diagram of a Hall Element

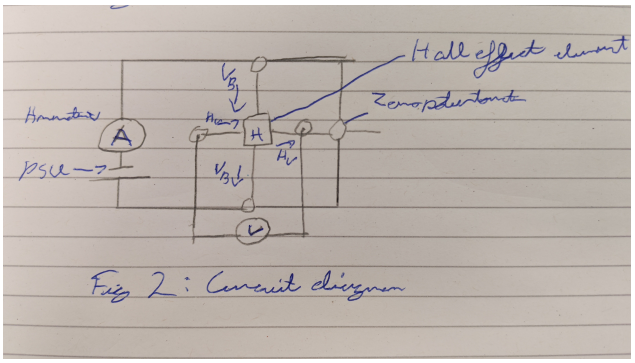


Figure 2: Circuit diagram of the apparatus

2. The sensor PSU was turned on and set to 0 V. The electromagnet PSU was set to 1 A.
3. The strength of the magnetic field inline with the sensor was recorded using a gaussmeter.
4. The hall sensor PSU was set to 4 mA. This value along with the voltage across the Hall sensor was recorded.
5. This was repeated 10 times until the amperage reached 40 mV.
6. The electomagnet PSU was set to 2 A and the previous measurments repeated. The strenght of the magnetic field was measured.
7. The PSU was then set to 3 A and the measurements were repeated again. The strenght of the magnetic field was measured.

3 Results

$$T = 1\,\text{mm}$$
$$R_H = \frac{V_H T}{IB}$$
$$R_{H(1\,\text{A})} = 0.8723\,\text{V}\,\text{A}^{-1} \cdot \frac{0.001\,\text{m}}{0.0739\,\text{T}}$$
$$R_{H(1\,\text{A})} = 1.18 \times 10^{-2}\,\text{C}$$

$$R_{H(2\,\text{A})} = 1.214\,\text{V}\,\text{A}^{-1} \cdot \frac{0.001\,\text{m}}{0.1215\,\text{T}}$$
$$R_{H(2\,\text{A})} = 9.92 \times 10^{-3}\,\text{C}$$

$$R_{H(3\,\text{A})} = 1.257\,\text{V}\,\text{A}^{-1} \cdot \frac{0.001\,\text{m}}{0.1505\,\text{T}}$$
$$R_{H(3\,\text{A})} = 8.352 \times 10^{-3}\,\text{C}$$

$$R_{H\ AVG} = \frac{1.18 \times 10^{-2}\,\text{C} + 9.92 \times 10^{-3}\,\text{C} + 8.352 \times 10^{-3}\,\text{C}}{3}$$
$$= 0.010\,04\,\text{C}$$
$$n = \frac{q}{R_H}$$
$$n = \frac{1}{1.602 \times 10^{-19}\,\text{C} \cdot 0.010\,04 \times 10^{-3}\,\text{C}} = 6.212 \times 10^{20}$$

4 Error Analysis

Multimeter error: 1%
Gaussmeter error: $\pm 0.01\,\text{mT}$
Electromagnet PSU error: $\pm 0.1\,\text{A}$

$$\Delta n = 6.212 \times 10^{20} \sqrt{0.01^2 + 0.05^2 + 0.01^2}$$
$$\Delta n = 3.2 \times 10^{19}$$

This error could have been reduced if the gaussmeter was placed in a stand of it's own, allowing consistant, accurate measurements.

5 Conclusion

The number of charge carriers within the hall effect sensor sucessfully determined by modulating the bias voltage while the sensor was held in a magnetic field. This value was found to be

$$n = (6.21 \pm 0.32) \times 10^{20}\ \text{Electrons}$$

6 References

1. Edwin Hall, 1879, *On a New Action of the Magnet on Electric Currents*, American Journal of Mathematics Vol. 2 No. 3

2. Manny Soltero, 2023, *What Is a Hall-effect Sensor?*, Texas Instruments technical article

3. Wei, Rongshan & Guo, Shizhong & Yang, Shanzhi, 2017, *Low-Power CMOS Integrated Hall Switch Sensor*, Active and Passive Electronic Components. 2017. 1-9. 10.1155/2017/5375619.